CGS616 ASSIGNMENT–2

Modeling the Temporal Dynamics of Latent Cognitive States Using Probabilistic and Learning-Based Frameworks

Due date: Feb 22, 2026

Background

In Assignment-1, you identified:

- Antecedents (A)
- Latent cognitive states (S)
- Observable behaviors (B)
- Consequences (C)

In Assignment-2, you will construct a **dynamic computational model** that explains how latent cognitive states evolve over time and generate observable behaviors.

Each Individual must:

1. Define a time-dependent latent state model for the chosen antecedent(s)
2. Estimate parameters from real data
3. Simulate behavioral trajectories
4. Evaluate goodness of fit
5. Interpret parameters psychologically

Each Team must compile from individuals the following, this can be reported in each individual report:

- Log-likelihood of individually picked antecedent or a combination of antecedent input driven models
- AIC / BIC of individual models (evaluation)
- Cross-validation accuracy (bonus)

1. Concept

Let:

$$\begin{aligned} S_t &= \text{latent cognitive state at time } t \\ X_t &= \text{stimulus/input (antecedent exposure)} \\ B_t &= \text{observable behavior} \\ \epsilon_t &= \text{noise} \end{aligned}$$

Your task is to model:

$$P(S_t \mid S_{t-1}, X_t)$$

and

$$P(B_t \mid S_t)$$

2. Model Family: Probabilistic State Transition Model (Markov / HMM)

State Transition Equation

For binary state $S_t \in \{0, 1\}$:

$$P(S_{t+1} = 1 \mid S_t, X_t) = \sigma(\beta X_t - \delta S_t + \epsilon_t)$$

That is,

$$\Delta S = f(\beta X_t - \delta S_t + \epsilon_t)$$

Where:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- δ = decay or recovery parameter
- β = sensitivity to stimulus
- η = Noise Parameter

$$\epsilon_t \sim \mathcal{N}(0, \tau^2)$$

Parameter Interpretation

Transition Probability:

$$P(S_t \mid S_{t-1})$$

Indicates stability vs. volatility of states.

Sensitivity to Stimulus (β): High $\beta \rightarrow$ rapid escalation Low $\beta \rightarrow$ resistant state
Decay/Recovery Rate (δ): High $\delta \rightarrow$ fast recovery/decay Low $\delta \rightarrow$ persistent state
Equilibrium Condition

$$S^* = \frac{\beta}{\delta} X$$

3a.You may consider updates using Reinforcement Learning

Q-Learning Update Rule

$$Q_{t+1} = Q_t + \alpha(r_t - Q_t)$$

- α = learning rate
- r_t = reward (likes, emotional arousal)
- Q_t = expected value of behavior

Behavior Probability (Softmax)

$$P(B_t) = \frac{e^{Q_t/\tau}}{\sum e^{Q_t/\tau}}$$

- τ = temperature (noise/exploration)

Parameter Interpretation

Learning Rate (α): High $\alpha \rightarrow$ rapid habit formation Low $\alpha \rightarrow$ gradual reinforcement

Reward Strength (r): Magnitude of reinforcement signal.

Temperature (τ): Low $\tau \rightarrow$ deterministic (exploitative) behavior High $\tau \rightarrow$ noisy behavior

3b.You may also consider using neural networks for state input approximation

4. Required Parameter Set

Each individual must report/estimate at least:

1. Transition probabilities
2. Parameter-Sensitivity to stimulus (β)
3. Parameter-decay / recovery rate (δ)
4. Parameter-learning rate (α if applicable)
5. Noise parameter (τ)

5. Optimization Procedure for parameters

Define likelihood:

$$L(\theta) = \prod P(B_t \mid S_t(\theta))$$

Estimate parameters:

$$\hat{\theta} = \arg\max L(\theta)$$

Or minimize:

$$\text{Loss} = -\log L(\theta)$$

Possible methods:

- Gradient Descent

- EM Algorithm (HMM)
- Maximum Likelihood Estimation (MLE)